



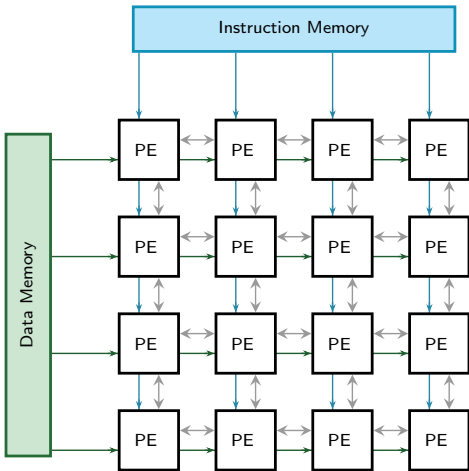
GNN-Guided Graph Partitioning for Spatial CGRA Mapping

Capstone Project Final Presentation

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Problem



DFG-to-CGRA mapping

- ▶ Place graph operations on a limited grid of processing elements.
- ▶ Respect operation compatibility, topology, and resource constraints.
- ▶ Split larger graphs into feasible spatial configurations.

The hard part is not only finding a placement, but choosing partitions that are likely to be mappable.

Direction

Learn

- ▶ LISA showed that GNNs can predict useful graph labels.

Validate

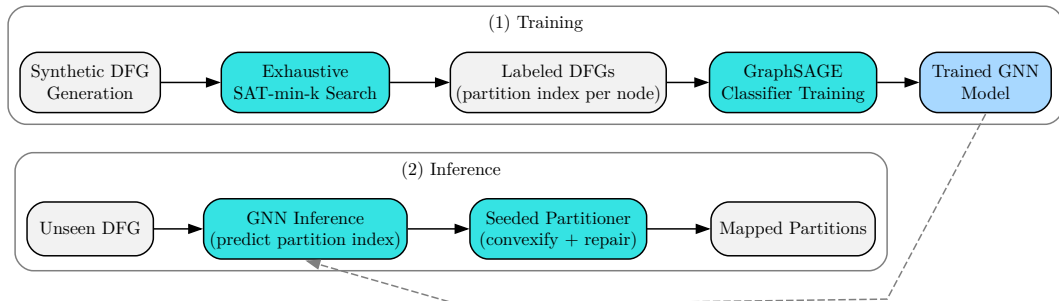
- ▶ SAT-MapIt motivated exact constraint checking.

Adapt

- ▶ The final prototype uses one-shot spatial partitions.

Use the GNN to seed the partitioning decision; use SAT to verify that each resulting partition can really be mapped.

Implemented Flow



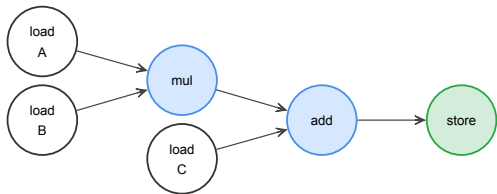
Training

- ▶ Generate synthetic DFGs.
- ▶ Search labels with exhaustive SAT-min-k.
- ▶ Train a GraphSAGE classifier.

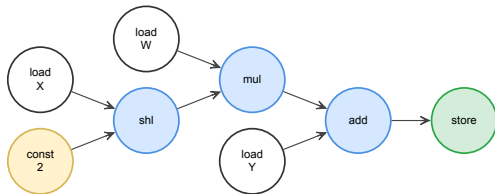
Inference

- ▶ Predict partition indices.
- ▶ Repair infeasible predictions.
- ▶ SAT-map each final partition.

Data and Ground Truth



Matrix multiplication template



Convolution template

- ▶ 1650 synthetic DFGs generated from kernel templates.
- ▶ 1500 used for training and ground-truth generation; 150 held out.
- ▶ Labels selected by feasible partitions with minimum cuts.

Evaluation

Strategy	Success	Mean cuts	Runtime
SAT-min-k	100.0%	1.31	12.1 s
Greedy	15.3%	1.52 [†]	0.08 s
Topological repair	92.7%	3.22	0.51 s
GNN-raw	70.7%	1.74	0.07 s
GNN-guided	98.0%	2.15	0.25 s

[†]Greedy cuts are reported only on its successful cases. Runtime is partitioning plus SAT mapping.

GNN-guided repair reached 147/150 mapped graphs and reduced cuts relative to unguided topological repair, while staying about 50× faster than exhaustive search.

Takeaway

Main contribution

- ▶ A working DFG-to-CGRA mapping pipeline.
- ▶ SAT-generated supervision for a learned partitioner.
- ▶ Feasibility repair tied to CGRA topology.

Current scope

- ▶ Homogeneous 4×4 CGRA.
- ▶ Synthetic DFGs.
- ▶ One-shot spatial mapping.

The result is not a replacement for exact mapping; it is a faster way to guide partitioning before exact validation.